

Ai Google Calendar Agent

Project title : AI-Powered Google Calendar Assistant

Basic idea,

- "Show my calendars"
- "What events do I have tomorrow?"
- "Add a meeting with Rahul tomorrow at 3 PM for one hour."
- "Change the meeting with Rahul to 5 PM."
- "Delete my 3 PM meeting"

Technology stack

Layer	Technology
Frontend	Chainlit or streamlit
AI	OpenAI API
Calendar	Google Calendar API
Authentication	Google OAuth 2.0
Language	python
Version Control	Git + Github

Main features

1. 📅 List calendar
2. 📋 List event
3. ➕ Add event
4. ✎ Edit event
5. Delete event

PROJECT STARTS HERE STEP BY STEP

Step 1. Create a file [Ai google calendar assistant](#), then create and activate venv, using: `python -m venv .venv`, `.venv\Scripts\activate`

Step 2. Install the packages,

- `pip update : python -m pip install --upgrade pip`
- **Google Calendar packages install: `pip install google-api-python-client google-auth-httplib2 google-auth-oauthlib`**
- `chainlit install : pip install chainlit`
- **OpenAI package install : `pip install openai`**

Step 3.create the text file [requirement.txt](#) it contains

- `chainlit`
- `Open ai`
- `google-api-python-client`
- `google-auth-httplib2`
- `google-auth-oauthlib`

step 4.crate app.py for testing

testing code: import chainlit as cl

@cl.on_message

async def main(message: cl.Message):

```
    await cl.Message(
        content=f"You said: {message.content}"
    ).send()
```

Command to run : python -m chainlit run app.py --port 8001

Maybe An error occur (reason): I used the python version 14 but that not supported the proper running of the chainlit operations so I changed the python version 14 to 13 (Chainlit 2.12.0 not support Python 3.14)
use this link for download python 3.13 : https://www.python.org/downloads/?utm_source=chatgpt.com

Deactivate the old venv and create a new venv (commands below to remove old venv)

- Deactivate
- Remove-Item -Recurse -Force .venv
- Then also install the packages shows the above

Step 5 : crate a .env file and store the open ai key

Store like , OPENAI_API_KEY=your api key here

Doubt How to get open ai key..?

Use this link https://platform.openai.com/api-keys?utm_source=chatgpt.com and click the option create new secret key option and create keys then copy and store to the .env folder

Step 6. We have to set the authentication use google cloud console to authenticate

Google clous console link :

https://console.cloud.google.com/welcome/new?utm_source=chatgpt.com&project=esoteric-drvad-508818-b3

- New Project → Project Name
- APIs & Services → Library ,then search google calendar api then enable it
- OAuth credentials create : APIs & Services → Credentials
- Download lient_secret.json
- Then crate a file credentials and store the .json file to it

Step 7. Create google_auth.py folder for complete authentication use this code

```
from google_auth_oauthlib.flow import InstalledAppFlow
```

```
from googleapiclient.discovery import build
```

```
from google.oauth2.credentials import Credentials
```

```
from google.auth.transport.requests import Request
```

```
import os
```

```
SCOPES = ["https://www.googleapis.com/auth/calendar"]
```

```
def get_calendar_service():
```

```
    creds = None
```

```
    # Check whether token already exists
```

```

if os.path.exists("token.json"):

    creds = Credentials.from_authorized_user_file(

        "token.json",

        SCOPES

    )

# If credentials are not valid, login again

if not creds or not creds.valid:

    if creds and creds.expired and creds.refresh_token:

        creds.refresh(Request())

    else:

        flow = InstalledAppFlow.from_client_secrets_file(

            "credentials/client_secret.json",

            SCOPES

        )

        creds = flow.run_local_server(port=0)

# Save login information

with open("token.json", "w") as token:

    token.write(creds.to_json())

# Create Google Calendar API service

service = build(

    "calendar",

    "v3",

    credentials=creds

)

return service

```

google gen ai api keys : https://aistudio.google.com/api-keys?utm_source=chatgpt.com&project=gen-lang-client-0679507369